

Sales Forecasting Glossary & Reference Guide

A

AutoETS ensemble: A computer algorithm that automatically selects the best exponential smoothing formula for each item's sales pattern and then averages multiple versions to get a more reliable prediction.

B

Base forecast: The initial predicted sales number produced directly by the model before any hierarchy adjustments are applied.

Bottom-up aggregation: Adding up all individual item forecasts to arrive at a total, rather than forecasting the total directly.

C

Chronos: A large AI model built by Amazon and pre-trained on millions of time series datasets that can forecast sales without being retrained on your specific data.

Coefficient of variation (CV): A measure of how erratic an item's sales are relative to its average, where a higher number means less predictable sales.

Confidence grade: A letter grade (A through D) summarizing how trustworthy the forecast for a given item is, based on past accuracy.

Confidence score: A number from 0 to 100 indicating how reliable the forecast for an item is, derived from how well the model performed historically on that item.

CrostonSBA: A specialized forecasting method for items that only sell occasionally, built to correct the tendency of standard methods to overpredict rare sales events.

Cross-series signal: Information borrowed from related products or the total market that helps the model predict a single item's future sales better.

Cyclical encoding (sin/cos): A technique that converts the month of the year into two smooth mathematical waves so the model understands that December and January are close together on the calendar, not far apart.

D

Damped trend: A forecasting technique where the model assumes that any current growth or decline will gradually slow down and flatten rather than continuing at the same rate forever.

Diebold-Mariano test: A statistical test that determines whether one forecasting method is genuinely more accurate than another or just got lucky on a particular dataset.

E

Ensemble: A strategy that combines the predictions of multiple different models into one final estimate, typically more accurate than any single model alone.

Expanding window: A validation setup where the training dataset grows larger with each test fold rather than staying a fixed size, mimicking how forecasting works in real life.

EWM (Exponential Weighted Mean): An average of past sales that gives more weight to recent months and gradually less weight to older ones.

F

Feature engineering: The process of constructing useful input variables (such as rolling averages, seasonal indicators, and price signals) that help the model recognize patterns.

Foundation model: A large AI model pre-trained on enormous amounts of data that can be applied to many different problems without being fully retrained from scratch.

G

Global model: A single machine learning model trained on all products simultaneously rather than fitting a separate model for each individual item.

H

Hierarchical reconciliation: A process that adjusts individual item forecasts so they add up correctly to match forecasts at the product-line level and at the total-company level.

Holt-Winters (Exponential Smoothing): A classic forecasting method that learns and accounts for both the overall direction of sales (trend) and predictable repeating patterns like seasonal peaks.

Hyperparameter tuning: The automated process of testing thousands of different model settings to find the combination that produces the most accurate forecasts.

I

Implicit price: An estimated unit price calculated by dividing total revenue by the number of units sold, used as a feature to capture pricing changes over time.

In-sample residuals: The differences between what the model predicted and what actually happened during the same time period the model was trained on.

Intermittent demand: A sales pattern where an item has zero sales in many months and only sells occasionally, making it much harder to forecast than items that sell consistently every month.

L

Lag features: Input variables created from past sales values (for example, sales from 3 or 12 months ago) that help the model learn how history predicts the future.

LightGBM: A fast and highly accurate machine learning algorithm developed by Microsoft that learns from patterns in historical data to make predictions.

M

M5-style features: A collection of input variables inspired by a well-known international forecasting competition, including rolling averages, seasonal signals, lagged quantities, and price estimates.

MAPE (Mean Absolute Percentage Error): A measure of forecast accuracy expressed as the average percentage by which predictions were off from actual sales.

MinT reconciliation (Minimum Trace): A mathematically optimal method for adjusting forecasts across all levels of a business hierarchy so they are internally consistent and collectively as accurate as possible.

Multi-resolution rolling: Computing averages of past sales over several different time window lengths at the same time (such as 2, 3, 6, 9, 12, and 24 months) to capture both short-term and long-term patterns simultaneously.

O

Optuna: An automated optimization library that searches through large numbers of possible model configurations to find the settings that minimize forecast error.

P

p-value: A number expressing how likely it would be to observe results this strong purely by chance, where a very small value (like 0.001) means the finding is statistically meaningful and not just luck.

Product-line lags: Recent sales figures for an entire group of related products used as an input signal to help predict one individual item's future sales.

Proportional scaling: A simpler reconciliation method used in the prior version of the model that scales item forecasts up or down by the same percentage to force them to match a known total.

R

Reconciliation: The step that adjusts forecasts at the item, product-line, and company-wide levels so they are all mathematically consistent with one another rather than summing to the wrong total.

Recursive forecasting: A process where the model predicts one month at a time and then feeds that prediction back in as an input when forecasting the following month.

Revenue-weighted MAPE: A version of forecast error where large-revenue items count more heavily toward the overall accuracy score than small low-revenue items.

Rolling window: A statistic (like an average or standard deviation) computed over a block of recent months that shifts forward one month at a time as new data arrives.

S

Seasonal naive: A simple baseline forecast that predicts the same value recorded in the same month one year earlier, used as the comparison point to show how much the model improves over doing nothing clever.

SKU (Stock Keeping Unit): A unique identifier code for one specific product variant, used to track sales and inventory at the most detailed level.

T

Tier: A revenue-based category grouping items from highest average monthly sales (Tier AA, above \$500K) down to the lowest tail items (Tier E, below \$1K), used to set accuracy expectations and choose appropriate modeling approaches.

Train-validation-test split: The practice of dividing historical data into three separate time periods: one to train the model, one to tune its settings, and one final holdout period for an unbiased evaluation of real performance.

Trend slope: A measure of how much an item's monthly sales are growing or shrinking on average over the training period, expressed as a fraction of average revenue.

W

Walk-forward cross-validation (CV): A testing method that simulates real-world forecasting by repeatedly training on all data up to a given point and then measuring how accurately the model predicted the next period.

Wilcoxon signed-rank test: A statistical test that checks whether one model's errors are systematically smaller than another's across all observations, without assuming the errors follow a bell-curve distribution.

WLS diagonal (Weighted Least Squares diagonal): The specific version of MinT reconciliation used here, where each item's adjustment is weighted by how variable its own historical forecast errors have been.

X

XGBoost: A machine learning algorithm similar to LightGBM that was used as the main forecasting model in the previous version of this system and serves as a performance comparison baseline.